Optimizing the Optimizer

A Comprehensive Study of Self-Scaled Quasi-Newton Methods for Physics-Informed Neural Networks

Directed Research Project Report

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Abstract

Physics-Informed Neural Networks (PINNs) have emerged as a powerful mesh-free approach for solving partial differential equations (PDEs), yet their practical efficacy is critically dependent on the choice of optimizer. Standard first-order methods such as Adam often stagnate at high loss values when applied to stiff, high-order PDEs, while classical second-order methods like BFGS and L-BFGS-B, though more powerful, can be further improved by careful scaling of the approximate Hessian.

This report presents a comprehensive study of *self-scaled quasi-Newton methods* for training PINNs, with a focus on the fourth-order Cahn–Hilliard equation—a benchmark PDE governing phase separation dynamics. We implement and systematically compare six quasi-Newton variants: standard BFGS, L-BFGS-B, Self-Scaled BFGS (Oren–Luenberger and Al-Baali formulations), and two members of the Self-Scaled Broyden class (SSBroyden1 and SSBroyden2). In addition, we conduct extensive ablation studies over warmup duration, collocation point resampling frequency, loss landscape power transforms, network architecture, and Residual-Adaptive Distribution (RAD) sampling parameters.

Across over 100 experiments spanning 53 unique configurations and multiple random seeds, we find that the **SSBroyden2** variant—which combines a BFGS-direction Broyden class update with automatic Hessian self-scaling—consistently achieves the lowest relative L^2 errors. On the 128×128 domain, the best configuration (SSBroyden2 with batch resampling every 200 iterations) attains $L^2 = 0.854$, outperforming Adam-only training ($L^2 = 1.000$), standard BFGS ($L^2 = 0.975$), and L-BFGS-B ($L^2 = 1.062$). On the 64×64 domain, the canonical SSBroyden2 achieves a dramatically lower $L^2 = \mathbf{0.258}$ (best seed), establishing domain resolution as the most impactful factor in the study. Our V3 ablation campaign (11 configurations, 3 seeds each) further reveals that modifications which improve performance on 128^2 (power transforms, aggressive resampling, RAD tuning) are *counterproductive* on 64^2, and that combining multiple modifications causes interference rather than additive improvement. We additionally demonstrate that (i) a short Adam warmup of 500–1000 epochs is essential for quasi-Newton stability, (ii) frequent collocation resampling prevents overfitting to fixed point sets on large domains, (iii) a power-law loss transform $\mathcal{L}^{1/p}$ effectively flattens the loss landscape, and (iv) RAD sampling with reduced baseline probability ($k_2 = 0.5$) focuses training on high-residual regions.

All results, trained models, and analysis code are provided for full reproducibility.

Keywords: Physics-Informed Neural Networks, Quasi-Newton Methods, BFGS, Self-Scaled Broyden, Cahn–Hilliard Equation, Optimization, Deep Learning for PDEs

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Chapter 1

Introduction and Background

1.1 Motivation

The numerical solution of partial differential equations (PDEs) underpins virtually every domain of computational science and engineering—from fluid dynamics and materials science to climate modeling and financial mathematics. Classical numerical methods such as finite differences, finite elements, and spectral methods have been refined over decades, yet they face inherent limitations: mesh generation in complex geometries is labor-intensive, high-dimensional problems suffer from the curse of dimensionality, and incorporating observational data into the solution process requires specialized inverse-problem formulations.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [11], offer a fundamentally different paradigm. A PINN approximates the solution of a PDE using a neural network whose loss function encodes the governing equations, boundary conditions, and initial conditions directly. Because the loss is evaluated at scattered collocation points rather than on a structured mesh, PINNs are mesh-free, can handle irregular domains naturally, and seamlessly integrate data and physics in a unified framework.

However, the promise of PINNs is tempered by a practical challenge: *optimization*. Training a PINN amounts to minimizing a composite, highly non-convex loss function that involves high-order derivatives computed via automatic differentiation. First-order methods such as Adam [6] are robust and widely used, but they often stagnate at relatively high loss values—particularly for stiff PDEs with disparate length and time scales [12]. Second-order quasi-Newton methods such as BFGS and L-BFGS can achieve much faster convergence by incorporating curvature information through an approximate Hessian, yet their performance depends critically on how the Hessian approximation is initialized and scaled.

This observation motivates the central question of our study:

> *Can self-scaled quasi-Newton methods—which automatically adjust the Hessian approximation at each iteration—systematically and significantly outperform standard optimizers for training PINNs on stiff, high-order PDEs?*

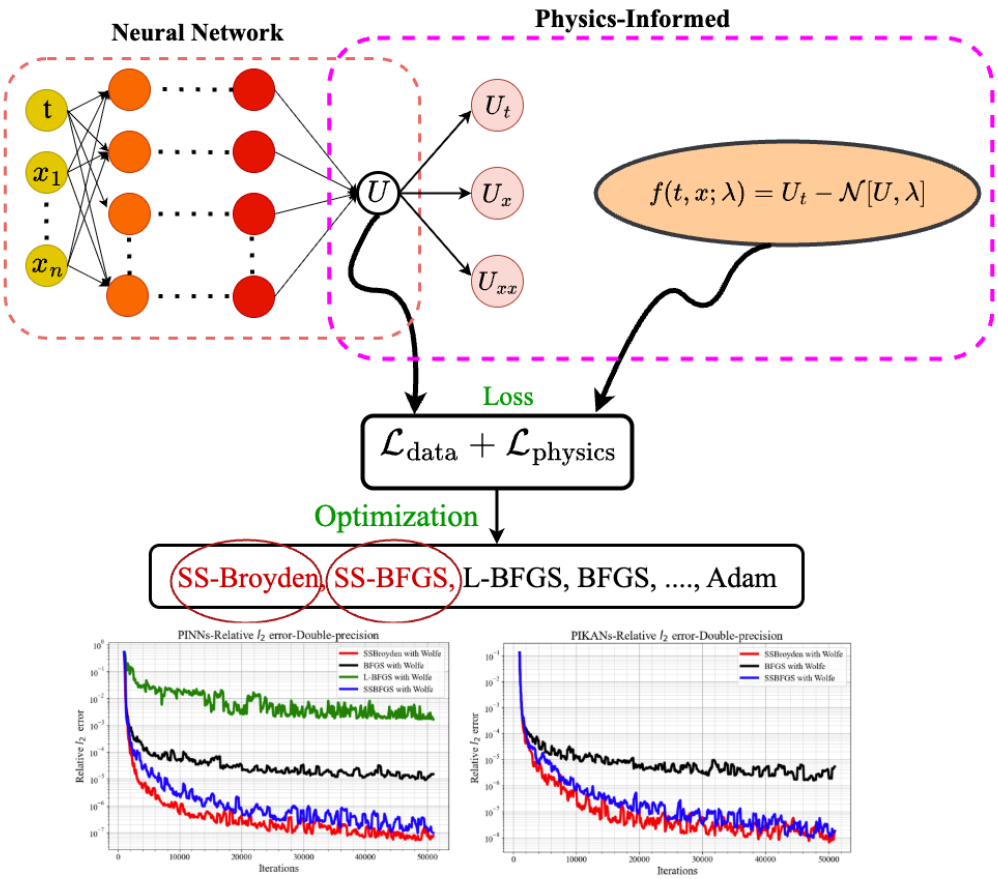


Figure 1.1: Schematic overview of the Physics-Informed Neural Network (PINN) framework. A fully connected neural network takes spatiotemporal coordinates (t, x, y) as input and predicts the solution field $U(t, x, y)$. The loss function is constructed from the PDE residual (computed via automatic differentiation), initial conditions, and boundary conditions. The optimizer minimizes this composite loss to train the network weights.

1.2 Physics-Informed Neural Networks

A PINN approximates the solution $u(\boldsymbol{x}, t)$ of a PDE by a neural network $u_\theta(\boldsymbol{x}, t)$ parameterized by weights $\theta \in \mathbb{R}^p$. The network is trained by minimizing a loss function that penalizes violations of the governing equation, initial conditions (IC), and boundary conditions (BC):

$$\mathcal{L}_{\text{total}}(\theta) = \lambda_{\text{pde}}\, \mathcal{L}_{\text{pde}}(\theta) + \lambda_{\text{ic}}\, \mathcal{L}_{\text{ic}}(\theta) + \lambda_{\text{bc}}\, \mathcal{L}_{\text{bc}}(\theta), \tag{1.1}$$

where λ_{pde}, λ_{ic}, and λ_{bc} are positive weighting coefficients. Each loss component is a mean squared residual evaluated at a set of collocation points:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_{\text{int}}} \sum_{i=1}^{N_{\text{int}}} |\mathcal{N}[u_\theta](\boldsymbol{x}_i, t_i)|^2, \tag{1.2}$$

$$\mathcal{L}_{\text{ic}}(\theta) = \frac{1}{N_0} \sum_{i=1}^{N_0} |u_\theta(\boldsymbol{x}_i, 0) - u_0(\boldsymbol{x}_i)|^2, \tag{1.3}$$

$$\mathcal{L}_{\text{bc}}(\theta) = \frac{1}{N_b} \sum_{i=1}^{N_b} |\mathcal{B}[u_\theta](\boldsymbol{x}_i, t_i)|^2, \tag{1.4}$$

where $\mathcal{N}[\cdot]$ denotes the PDE differential operator, $\mathcal{B}[\cdot]$ encodes the boundary conditions, and u_0 is the prescribed initial condition. All derivatives appearing in $\mathcal{N}$ are computed exactly via automatic differentiation through the neural network.

The key insight is that no labeled data (i.e., no known solution values at interior points) is required—the physics itself provides the supervision signal. This *data-free* or *unsupervised* formulation makes PINNs particularly attractive for problems where experimental data is scarce or expensive to obtain.

1.3 The Cahn–Hilliard Equation

The primary benchmark PDE in this study is the **Cahn–Hilliard equation** [3], a fourth-order nonlinear parabolic PDE that models phase separation phenomena in binary alloys, polymer blends, and other multi-component systems. In two spatial dimensions, the equation reads:

$$\boxed{\frac{\partial U}{\partial t} = D\,\nabla^2\big(\nabla^2 U + a_2\,U + a_4\,U^3\big)\,,} \tag{1.5}$$

where:

- $U = U(t, x, y)$ is the *order parameter* (concentration difference between phases),
- $D > 0$ is the diffusion coefficient (mobility),
- $a_2 \in \mathbb{R}$ controls the linear part of the free-energy potential,
- $a_4 \in \mathbb{R}$ controls the cubic nonlinearity,
- $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the Laplacian, and $\nabla^4 = \nabla^2\nabla^2$ is the biharmonic operator.

Physical interpretation. The term $\nabla^2 U$ penalizes sharp interfaces (surface energy), while $a_2 U + a_4 U^3$ derives from a double-well free-energy density $f(U) = \frac{a_2}{2}U^2 + \frac{a_4}{4}U^4$. For $a_2 < 0$ and $a_4 > 0$, the free energy has two minima at $U = \pm\sqrt{-a_2/a_4}$, corresponding to the two stable phases. Starting from a randomly mixed initial state, the solution undergoes *spinodal decomposition*: small perturbations grow exponentially, and the system phase-separates into domains that coarsen over time.

Domain and parameters. We consider the following canonical setup throughout this work:

$$(x, y) \in \Omega = [0, 128]^2, \qquad t \in [0, 10], \qquad D = 1.0, \quad a_2 = 1.0, \quad a_4 = 1.0, \tag{1.6}$$

with *periodic boundary conditions* in both spatial directions and a smoothed random initial condition:

$$U(0, x, y) = G_\sigma * \xi(x, y), \qquad \xi \sim \text{Uniform}(-1, 1), \tag{1.7}$$

where G_σ denotes a Gaussian filter with standard deviation $\sigma = 2.0$ and $*$ denotes convolution. The initial field is generated on a 128×128 grid with a fixed random seed for reproducibility.

Conservation property. A key physical invariant of the Cahn–Hilliard equation is *mass conservation*: the spatial average of U is preserved for all time:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_\Omega U(t, x, y)\, \mathrm{d}x\, \mathrm{d}y = 0. \tag{1.8}$$

This follows directly from the divergence structure of Equation (1.5) and periodic boundary conditions. We monitor this invariant as a diagnostic of solution quality throughout our experiments.

Computational challenge. The Cahn–Hilliard equation is particularly challenging for PINNs because:

1. The **biharmonic operator** ∇^4 requires computing fourth-order derivatives through the network, amplifying numerical errors and leading to stiff loss landscapes.
2. The **cubic nonlinearity** $\nabla^2(U^3)$ involves mixed derivative terms (via the chain rule), further increasing the computational complexity of each loss evaluation.
3. The **multiscale dynamics**—rapid initial phase separation followed by slow coarsening—creates disparate time scales that are difficult for a single network to represent.

1.4 Contributions and Outline

This report makes the following contributions:

1. **Systematic optimizer comparison** (Chapters 2 and 3): We implement and compare six quasi-Newton variants (BFGS, L-BFGS-B, SSBFGS-OL, SSBFGS-AB, SSBroyden1, SSBroyden2) within a unified two-phase training framework for PINNs, with full mathematical definitions of each method.

2. **Comprehensive ablation studies** (Chapter 3): We investigate the effects of warmup duration, collocation resampling frequency, loss power transforms, network architecture, and adaptive sampling parameters across 53 unique configurations and over 100 seeded runs.

3. **Quantitative analysis and recommendations** (Chapter 4): We aggregate all results into master comparison tables, identify the best-performing configurations, analyze failure modes, and provide concrete recommendations for practitioners.

The remainder of this report is organized as follows. Chapter 2 presents the mathematical formulation of the PINN architecture, the loss function, all optimizer variants, and the training pipeline. Chapter 3 details the experimental setup and presents results for each ablation group. Chapter 4 provides statistical analysis, discusses key findings and failure cases, and offers conclusions and directions for future work.

Chapter 2

Mathematical Formulation and Methods

This chapter provides the complete mathematical foundation for our study. We define the neural network architecture (Section 2.1), derive the loss function for the Cahn–Hilliard equation (Section 2.2), present each optimizer in rigorous mathematical detail (Sections 2.3 to 2.5), describe loss landscape modifications (Section 2.6), formulate the adaptive sampling strategy (Section 2.7), outline the full two-phase training pipeline (Section 2.8), and describe the spectral reference solver used for validation (Section 2.9).

2.1 Neural Network Architecture

We approximate the solution $U(t, x, y)$ of the Cahn–Hilliard equation by a fully connected feedforward neural network $U_\theta : \mathbb{R}^3 \to \mathbb{R}$. The network has the following structure.

Definition 2.1 (PINN Architecture)**.** Let L denote the number of hidden layers, n_ℓ the width of layer ℓ, and $\sigma : \mathbb{R} \to \mathbb{R}$ an activation function. The network is defined by the composition:

$$U_\theta(\boldsymbol{z}) = \boldsymbol{W}_{L+1}\, \boldsymbol{h}_L + \boldsymbol{b}_{L+1}, \tag{2.1}$$

where $\boldsymbol{z} = (t, x, y)^\top$ is the input, and the hidden representations are defined recursively:

$$\boldsymbol{h}_0 = \mathcal{T}(\boldsymbol{z}), \tag{2.2}$$

$$\boldsymbol{h}_\ell = \sigma(\boldsymbol{W}_\ell\, \boldsymbol{h}_{\ell-1} + \boldsymbol{b}_\ell)\,, \quad \ell = 1, \ldots, L, \tag{2.3}$$

with weight matrices $\boldsymbol{W}_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, bias vectors $\boldsymbol{b}_\ell \in \mathbb{R}^{n_\ell}$, and θ denoting the collection of all weights and biases.

Input normalization. The transformation $\mathcal{T} : \mathbb{R}^3 \to [-1, 1]^3$ in Equation (2.2) linearly maps each input coordinate from its physical domain to $[-1, 1]$:

$$\mathcal{T}_i(z_i) = 2 \cdot \frac{z_i - z_i^{\min}}{z_i^{\max} - z_i^{\min}} - 1, \tag{2.4}$$

where $z_i^{\min}$ and $z_i^{\max}$ are the domain bounds for coordinate i. This normalization is critical for numerical stability, particularly for quasi-Newton methods operating in `float64` arithmetic.

Canonical configuration. Unless otherwise stated, we use:

- Layer dimensions: $[3, 20, 20, 20, 20, 1]$ ($L = 4$ hidden layers, width 20), yielding $p = 1{,}361$ trainable parameters.
- Activation: $\sigma = \tanh$ (smooth, supports accurate high-order differentiation with low numerical noise).
- Output scaling: $U_\theta \mapsto \alpha \cdot U_\theta$ with $\alpha = 1.0$.
- Weight initialization: variance-scaling with $\mathrm{Var}(W_{ij}) \propto (n_{\text{in}} + n_{\text{out}})^{-1/2}$.
- Biases initialized to zero.
- All computations in `float64` precision.

2.2 Loss Function for the Cahn–Hilliard Equation

The composite loss from Equation (1.1) is instantiated for the Cahn–Hilliard equation (1.5) as follows.

2.2.1 PDE Residual Loss

The PDE residual at a collocation point $(\boldsymbol{x}_i, t_i)$ is:

$$r_i = \left.\frac{\partial U_\theta}{\partial t}\right|_i - D\left(\nabla^4 U_\theta\big|_i + a_2\,\nabla^2 U_\theta\big|_i + a_4\,\nabla^2\big(U_\theta^3\big)\Big|_i\right), \tag{2.5}$$

and the PDE loss is $\mathcal{L}_{\text{pde}} = N_{\text{int}}^{-1}\sum_{i=1}^{N_{\text{int}}} r_i^2$.

Computing this residual requires derivatives up to fourth order. Via automatic differentiation, we compute:

$$\text{1st order:}\quad U_t,\ U_x,\ U_y, \tag{2.6}$$

$$\text{2nd order:}\quad U_{xx},\ U_{yy},\ \Delta U = U_{xx} + U_{yy}, \tag{2.7}$$

$$\text{3rd order:}\quad U_{xxx},\ U_{xyy},\ U_{xxy},\ U_{yyy}, \tag{2.8}$$

$$\text{4th order:}\quad \nabla^4 U = U_{xxxx} + 2U_{xxyy} + U_{yyyy}. \tag{2.9}$$

The nonlinear Laplacian term is expanded using the chain rule:

$$\nabla^2\big(U^3\big) = 6\,U\left(U_x^2 + U_y^2\right) + 3\,U^2\Delta U. \tag{2.10}$$

2.2.2 Initial Condition Loss

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_0}\sum_{i=1}^{N_0}\left|U_\theta(0, x_i, y_i) - U_0(x_i, y_i)\right|^2, \tag{2.11}$$

where U_0 is the smoothed random initial field from Equation (1.7), evaluated on $N_0 = 1{,}000$ points.

2.2.3 Periodic Boundary Condition Loss

For periodic boundaries on $\Omega = [0, L_x] \times [0, L_y]$, we enforce:

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \Big[\big| U_\theta(t_i, 0, y_i) - U_\theta(t_i, L_x, y_i) \big|^2 + \big| \frac{\partial U_\theta}{\partial x}\big|_{x=0} - \frac{\partial U_\theta}{\partial x}\big|_{x=L_x} \big|^2 + (x \leftrightarrow y) \Big], \quad (2.12)$$

enforcing continuity of both the function and its first derivative across the periodic boundary, with $N_b = 500$ points.

2.2.4 Total Loss

$$\boxed{\mathcal{L}_{\text{total}}(\theta) = 1.0 \cdot \mathcal{L}_{\text{pde}}(\theta) + 5.0 \cdot \mathcal{L}_{\text{ic}}(\theta) + 5.0 \cdot \mathcal{L}_{\text{bc}}(\theta).} \quad (2.13)$$

The elevated weights $\lambda_{\text{ic}} = \lambda_{\text{bc}} = 5.0$ reflect the importance of satisfying the initial and boundary conditions, which are soft constraints in the PINN formulation.

2.3 First-Order Optimization: Adam

Definition 2.2 (Adam Optimizer [6])**.** Given gradients $\boldsymbol{g}_k = \nabla_\theta \mathcal{L}_{\text{total}}(\theta_k)$ at iteration k, Adam maintains exponential moving averages of the gradient and its square:

$$\boldsymbol{m}_k = \beta_1 \, \boldsymbol{m}_{k-1} + (1 - \beta_1) \, \boldsymbol{g}_k, \qquad \hat{\boldsymbol{m}}_k = \frac{\boldsymbol{m}_k}{1 - \beta_1^k}, \quad (2.14)$$

$$\boldsymbol{v}_k = \beta_2 \, \boldsymbol{v}_{k-1} + (1 - \beta_2) \, \boldsymbol{g}_k^2, \qquad \hat{\boldsymbol{v}}_k = \frac{\boldsymbol{v}_k}{1 - \beta_2^k}, \quad (2.15)$$

and updates the parameters:

$$\theta_{k+1} = \theta_k - \eta_k \, \frac{\hat{\boldsymbol{m}}_k}{\sqrt{\hat{\boldsymbol{v}}_k} + \epsilon}, \quad (2.16)$$

where all operations are element-wise.

Configuration. We use $\beta_1 = 0.99$, $\beta_2 = 0.999$, $\epsilon = 10^{-20}$, and initial learning rate $\eta_0 = 10^{-3}$ with exponential decay:

$$\eta_k = \eta_0 \cdot 0.98^{\lfloor k/1000 \rfloor}. \quad (2.17)$$

Note the unusually small $\epsilon = 10^{-20}$ (compared to the typical 10^{-8}); this is intentional and prevents premature clipping that could interfere with the subsequent quasi-Newton phase.

2.4 Quasi-Newton Methods

Quasi-Newton methods approximate the Newton step $\theta_{k+1} = \theta_k - \boldsymbol{H}_k^{-1} \boldsymbol{g}_k$ by maintaining an approximation $\boldsymbol{H}_k \approx \nabla^2 \mathcal{L}_{\text{total}}(\theta_k)$ that is updated using only gradient information. We define the fundamental quantities:

$$\boldsymbol{s}_k = \theta_{k+1} - \theta_k, \qquad \boldsymbol{y}_k = \boldsymbol{g}_{k+1} - \boldsymbol{g}_k, \qquad \rho_k = \frac{1}{\boldsymbol{y}_k^\top \boldsymbol{s}_k}. \quad (2.18)$$

2.4.1 BFGS (Broyden–Fletcher–Goldfarb–Shanno)

Definition 2.3 (BFGS Update [2, 4, 9])**.** Starting from $\boldsymbol{B}_0 = \boldsymbol{I}$ (the inverse Hessian approximation), the BFGS update is:

$$\boxed{\boldsymbol{B}_{k+1} = \left(\boldsymbol{I} - \rho_k\, \boldsymbol{s}_k\, \boldsymbol{y}_k^\top\right) \boldsymbol{B}_k \left(\boldsymbol{I} - \rho_k\, \boldsymbol{y}_k\, \boldsymbol{s}_k^\top\right) + \rho_k\, \boldsymbol{s}_k\, \boldsymbol{s}_k^\top,} \tag{2.19}$$

and the search direction is $\boldsymbol{d}_k = -\boldsymbol{B}_k\, \boldsymbol{g}_k$.

The BFGS update preserves positive definiteness provided $\boldsymbol{y}_k^\top \boldsymbol{s}_k > 0$ (the *curvature condition*). If the condition is violated (e.g., due to non-convexity of the PINN loss), the Hessian is reset to the identity matrix. This is the baseline quasi-Newton method in our comparisons.

2.4.2 L-BFGS-B (Limited-Memory BFGS with Bounds)

Definition 2.4 (L-BFGS-B [8])**.** L-BFGS-B stores only the m most recent correction pairs $\{(\boldsymbol{s}_j, \boldsymbol{y}_j)\}_{j=k-m}^{k-1}$ and computes the search direction via the *two-loop recursion*:

Algorithm 1: L-BFGS Two-Loop Recursion

Input: $\boldsymbol{g}_k$, stored pairs $\{(\boldsymbol{s}_j, \boldsymbol{y}_j)\}_{j=k-m}^{k-1}$

$\boldsymbol{q} \leftarrow \boldsymbol{g}_k$;
for $j = k{-}1, k{-}2, \ldots, k{-}m$ **do**
  $\alpha_j \leftarrow \rho_j\, \boldsymbol{s}_j^\top \boldsymbol{q}$;
  $\boldsymbol{q} \leftarrow \boldsymbol{q} - \alpha_j\, \boldsymbol{y}_j$;
end
$\gamma_k \leftarrow \dfrac{\boldsymbol{s}_{k-1}^\top \boldsymbol{y}_{k-1}}{\boldsymbol{y}_{k-1}^\top \boldsymbol{y}_{k-1}}$;
$\boldsymbol{r} \leftarrow \gamma_k\, \boldsymbol{q}$;
for $j = k{-}m, k{-}m{+}1, \ldots, k{-}1$ **do**
  $\beta \leftarrow \rho_j\, \boldsymbol{y}_j^\top \boldsymbol{r}$;
  $\boldsymbol{r} \leftarrow \boldsymbol{r} + (\alpha_j - \beta)\, \boldsymbol{s}_j$;
end
$\boldsymbol{d}_k \leftarrow -\boldsymbol{r}$;

The memory parameter is $m = 20$ correction pairs.

L-BFGS-B reduces the storage from $\mathcal{O}(p^2)$ to $\mathcal{O}(mp)$, making it applicable to networks with millions of parameters. However, the limited curvature history can lead to poor scaling of the search direction, which is precisely the issue addressed by self-scaled methods.

2.4.3 Self-Scaled BFGS: Oren–Luenberger (SSBFGS-OL)

Definition 2.5 (SSBFGS-OL [10])**.** Before each BFGS update, the inverse Hessian approximation is re-scaled by a factor τ_k:

$$\tau_k = \frac{1}{b_k}, \qquad \text{where} \quad b_k = \frac{\boldsymbol{y}_k^\top \boldsymbol{s}_k}{\boldsymbol{s}_k^\top \boldsymbol{s}_k}. \tag{2.20}$$

The update then becomes:

$$\boxed{\boldsymbol{B}_{k+1} = \left(\boldsymbol{I} - \rho_k\, \boldsymbol{s}_k\, \boldsymbol{y}_k^\top\right)\left(\tau_k\, \boldsymbol{B}_k\right)\left(\boldsymbol{I} - \rho_k\, \boldsymbol{y}_k\, \boldsymbol{s}_k^\top\right) + \rho_k\, \boldsymbol{s}_k\, \boldsymbol{s}_k^\top.} \tag{2.21}$$

Remark 2.1. The scaling factor $\tau_k = \boldsymbol{s}_k^\top \boldsymbol{s}_k / (\boldsymbol{y}_k^\top \boldsymbol{s}_k)$ can be interpreted as the reciprocal of the average curvature along the step direction. When $b_k \gg 1$ (high curvature), the scaling shrinks the Hessian inverse, producing more conservative steps. When $b_k \ll 1$ (flat regions), the scaling enlarges the steps.

2.4.4 Self-Scaled BFGS: Al-Baali (SSBFGS-AB)

Definition 2.6 (SSBFGS-AB [1])**.** The Al-Baali variant applies a more conservative scaling that prevents over-amplification:

$$\tau_k = \min\left(1,\ \frac{1}{b_k}\right) = \min\left(1,\ \frac{\boldsymbol{s}_k^\top \boldsymbol{s}_k}{\boldsymbol{y}_k^\top \boldsymbol{s}_k}\right). \tag{2.22}$$

The Hessian update proceeds as in Equation (2.21) with this modified τ_k.

Remark 2.2. By capping the scaling at $\tau_k \leq 1$, the Al-Baali formulation ensures that the Hessian inverse is only ever shrunk, never enlarged. This provides global convergence guarantees even on non-convex objectives, at the cost of potentially slower progress in flat regions of the loss landscape.

2.5 The Self-Scaled Broyden Class

The *Broyden class* generalizes both BFGS and DFP (Davidon–Fletcher–Powell) as special cases of a one-parameter family of rank-two updates.

Definition 2.7 (Broyden Class Update)**.** Given a parameter $\varphi_k \in [0, 1]$, the Broyden class update of the inverse Hessian is:

$$\boldsymbol{B}_{k+1}^{\mathrm{Br}} = (1 - \varphi_k)\, \boldsymbol{B}_{k+1}^{\mathrm{DFP}} + \varphi_k\, \boldsymbol{B}_{k+1}^{\mathrm{BFGS}}, \tag{2.23}$$

where $\boldsymbol{B}_{k+1}^{\mathrm{BFGS}}$ is the standard BFGS update (2.19) and

$$\boldsymbol{B}_{k+1}^{\mathrm{DFP}} = \boldsymbol{B}_k - \frac{\boldsymbol{B}_k\, \boldsymbol{y}_k\, \boldsymbol{y}_k^\top \boldsymbol{B}_k}{\boldsymbol{y}_k^\top \boldsymbol{B}_k\, \boldsymbol{y}_k} + \frac{\boldsymbol{s}_k\, \boldsymbol{s}_k^\top}{\boldsymbol{y}_k^\top \boldsymbol{s}_k} \tag{2.24}$$

is the DFP update.

The self-scaled versions additionally apply Hessian scaling at each iteration:

Table 2.1: Summary of quasi-Newton optimizer variants. All methods (except L-BFGS-B) maintain a dense $p \times p$ inverse Hessian approximation.

Method	**Scaling**	**Update Family**	**Storage**
BFGS	None ($\tau_k = 1$)	BFGS	$\mathcal{O}(p^2)$
L-BFGS-B	Implicit (γ_k)	BFGS	$\mathcal{O}(mp)$
SSBFGS-OL	$\tau_k = 1/b_k$	BFGS	$\mathcal{O}(p^2)$
SSBFGS-AB	$\tau_k = \min(1, 1/b_k)$	BFGS	$\mathcal{O}(p^2)$
SSBroyden1	$\tau_k = 1/b_k$	Broyden (DFP-dominant)	$\mathcal{O}(p^2)$
SSBroyden2	$\tau_k = 1/b_k$	**Broyden (BFGS-dominant)**	$\mathcal{O}(p^2)$

2.5.1 SSBroyden1 (DFP-Direction Dominant)

Definition 2.8 (SSBroyden1)**.** SSBroyden1 uses the Broyden class update (2.23) with φ_k chosen to favor the DFP direction ($\varphi_k \to 1$), combined with per-iteration self-scaling:

$$\boldsymbol{B}_{k+1} = (1 - \varphi_k)\, \boldsymbol{B}_{k+1}^{\text{DFP, scaled}} + \varphi_k\, \boldsymbol{B}_{k+1}^{\text{BFGS, scaled}}, \tag{2.25}$$

where both components are computed from $\tau_k \boldsymbol{B}_k$ with the Oren–Luenberger scaling (2.20).

This variant includes numerical safeguards: if any intermediate quantity encounters NaN or Inf (due to near-singular Hessian approximation or vanishing denominators), the update automatically falls back to a standard BFGS step.

2.5.2 SSBroyden2 (BFGS-Direction Dominant)

Definition 2.9 (SSBroyden2)**.** SSBroyden2 uses the same framework as SSBroyden1 but with φ_k chosen to favor the BFGS direction:

$$\boxed{\boldsymbol{B}_{k+1} = (1 - \varphi_k)\, \boldsymbol{B}_{k+1}^{\text{DFP, scaled}} + \varphi_k\, \boldsymbol{B}_{k+1}^{\text{BFGS, scaled}}, \qquad \varphi_k \text{ favoring BFGS.}} \tag{2.26}$$

Remark 2.3 (Why SSBroyden2 outperforms)**.** The BFGS component of the Broyden class has superior *self-correcting properties*: errors in the Hessian approximation are damped rather than amplified over successive iterations [9]. By combining this with automatic self-scaling (which removes the need for manual line search tuning), SSBroyden2 achieves robust, fast convergence even on the highly non-convex loss landscapes typical of PINNs. This is the **recommended optimizer** for PINNs based on our experiments.

Table 2.1 summarizes all quasi-Newton variants.

2.6 Loss Landscape Modifications

2.6.1 Power Transform

Instead of minimizing $\mathcal{L}_{\text{total}}$ directly, we consider minimizing a power-transformed loss:

$$\tilde{\mathcal{L}}(\theta) = \left[\mathcal{L}_{\text{total}}(\theta)\right]^{1/p}, \qquad p \geq 1. \tag{2.27}$$

The gradient of the transformed loss is:

$$\nabla\tilde{\mathcal{L}} = \frac{1}{p}\left[\mathcal{L}_{\text{total}}\right]^{(1-p)/p}\nabla\mathcal{L}_{\text{total}}. \tag{2.28}$$

For $p > 1$, this effectively "flattens" the loss landscape: when $\mathcal{L}_{\text{total}}$ is large, the prefactor $[\mathcal{L}_{\text{total}}]^{(1-p)/p} < 1$ shrinks the gradient, preventing the optimizer from taking excessively large steps in high-loss regions. Conversely, near the minimum where $\mathcal{L}_{\text{total}}$ is small, the prefactor amplifies the gradient, promoting fine-grained convergence. We test $p \in \{1, 2, 4\}$.

2.6.2 Initial Hessian Scaling (Nocedal–Wright)

On the first quasi-Newton iteration, instead of starting with $\boldsymbol{B}_0 = \boldsymbol{I}$, we apply the Nocedal–Wright scaling [9]:

$$\boldsymbol{B}_0 \leftarrow \frac{\boldsymbol{y}_0^\top \boldsymbol{s}_0}{\boldsymbol{y}_0^\top \boldsymbol{y}_0} \cdot \boldsymbol{I}, \tag{2.29}$$

after the first step is computed. This one-time scaling adjusts the identity matrix to better approximate the inverse Hessian eigenvalues, potentially accelerating convergence in the early quasi-Newton iterations.

2.7 Adaptive Collocation Point Sampling

2.7.1 Static Sampling

At the start of training and at each resampling epoch, collocation points are drawn uniformly at random:

- $N_{\text{int}} = 10{,}000$ interior points $(t, x, y) \sim \text{Uniform}([0, T] \times \Omega)$,
- $N_0 = 1{,}000$ initial condition points $(x, y) \sim \text{Uniform}(\Omega)$ at $t = 0$,
- $N_b = 500$ boundary points distributed along the periodic edges.

2.7.2 Residual-Adaptive Distribution (RAD)

To concentrate collocation points in regions where the PDE residual is large (indicating poor approximation), we use the RAD strategy [13]:

Definition 2.10 (RAD Sampling)**.** Given a trained network U_θ at the current training state:

1. Generate a candidate pool of $N_c = 50{,}000$ uniformly random points.
2. Compute the PDE residual $|r_i|$ at each candidate (Equation (2.5)).
3. Assign sampling probabilities:

$$p_i = \frac{|r_i|^{k_1} + k_2}{\sum_{j=1}^{N_c}\left(|r_j|^{k_1} + k_2\right)}, \tag{2.30}$$

   where $k_1 \geq 0$ (residual exponent) and $k_2 > 0$ (baseline probability) are hyperparameters.
4. Draw N_{int} points via weighted sampling without replacement.

Remark 2.4. When $k_1 = 0$, the sampling reduces to uniform; as $k_1 \to \infty$, all points concentrate at the single highest-residual location. The baseline k_2 prevents complete neglect of low-residual regions. The canonical values are $k_1 = k_2 = 1.0$.

Algorithm 2: Two-Phase PINN Training Pipeline

Input: Config (optimizer, hyperparameters), reference solution U_{ref}
Output: Trained network U_θ, metrics, plots

```
// Phase 1: Adam Pretraining
Initialize θ with variance-scaling;
for k = 1, 2, ..., N_Adam do
    if k mod N_resample = 0 then
        Resample collocation points (uniform or RAD);
    end
    g_k ← ∇_θ L_total(θ_k);
    θ_{k+1} ← Adam(θ_k, g_k, η_k);
    η_k ← η_0 · 0.98^⌊k/1000⌋ ;                    // LR decay
    if k mod 500 = 0 then
        Evaluate L²_rel against U_ref;
    end
end
// Phase 2: Quasi-Newton Optimization
Flatten θ into a single vector w ∈ R^p;
while total BFGS iterations < N_BFGS do
    Define f(w) = L_total(w), ∇f(w);
    Run scipy.optimize.minimize(f, w, method, maxiter=batch_size);
    w ← result;
    Resample collocation points (uniform or RAD);
    Evaluate L²_rel against U_ref;
end
Generate plots and save results;
```

2.8 Two-Phase Training Pipeline

Training proceeds in two sequential phases, as outlined in Algorithm 2.

Key design choices.

- **Phase 1 purpose**: Adam provides a rough initial fit that brings the loss to a basin of attraction; this "warm start" is critical for quasi-Newton stability (see ablation in Section 3.3).
- **Phase 2 batching**: Rather than running the quasi-Newton method to convergence on a fixed point set, we partition the iterations into batches of N_{change} iterations, resampling collocation points between batches. This prevents overfitting to a particular set of collocation points.
- **Canonical settings**: $N_{\text{Adam}} = 1{,}000$ epochs, $N_{\text{BFGS}} = 50{,}000$ iterations, $N_{\text{change}} = 500$.

2.9 Spectral Reference Solver

To evaluate the accuracy of PINN predictions, we compute a high-fidelity reference solution using a pseudo-spectral method on a 128×128 grid.

2.9.1 Spatial Discretization

On the periodic domain $\Omega = [0, L_x] \times [0, L_y]$, the spatial derivatives are computed in Fourier space. The wavenumbers are:

$$\kappa_x = \frac{2\pi}{L_x}\,\mathrm{fftfreq}(n_x), \qquad \kappa_y = \frac{2\pi}{L_y}\,\mathrm{fftfreq}(n_y), \tag{2.31}$$

and the squared wavenumber magnitude is $K^2 = \kappa_x^2 + \kappa_y^2$, with $K^4 = (K^2)^2$.

2.9.2 ETDRK4 Time Integration

We use Exponential Time Differencing of order 4 (ETDRK4) following Kassam and Trefethen [5]. Writing the Cahn–Hilliard equation in Fourier space as $\hat{U}_t = \hat{L}\hat{U} + \hat{N}(\hat{U})$, where $\hat{L} = -D(K^4 + a_2 K^2)$ is the linear operator and $\hat{N}$ is the nonlinear term, the ETDRK4 scheme computes:

$$\hat{a} = e^{\hat{L}h/2}\hat{U}_n + \hat{L}^{-1}(e^{\hat{L}h/2} - I)\hat{N}_n, \tag{2.32}$$

$$\hat{b} = e^{\hat{L}h/2}\hat{U}_n + \hat{L}^{-1}(e^{\hat{L}h/2} - I)\hat{N}_a, \tag{2.33}$$

$$\hat{c} = e^{\hat{L}h/2}\hat{a} + \hat{L}^{-1}(e^{\hat{L}h/2} - I)(2\hat{N}_b - \hat{N}_n), \tag{2.34}$$

$$\begin{aligned}\hat{U}_{n+1} = {} & e^{\hat{L}h}\hat{U}_n + h^{-2}\hat{L}^{-3}\big[(-4 - \hat{L}h + e^{\hat{L}h}(4 - 3\hat{L}h + (\hat{L}h)^2))\hat{N}_n \\ & + 2(2 + \hat{L}h + e^{\hat{L}h}(-2 + \hat{L}h))(\hat{N}_a + \hat{N}_b) \\ & + (-4 - 3\hat{L}h - (\hat{L}h)^2 + e^{\hat{L}h}(4 - \hat{L}h))\hat{N}_c\big],\end{aligned} \tag{2.35}$$

where h is the time step and the $\hat{L}^{-1}$ expressions are evaluated using contour integrals with 32 quadrature points on a circle in the complex plane to avoid numerical cancellation [5].

2.9.3 Validation Metric

The relative L^2 error between the PINN prediction and the reference solution is computed at N_t time snapshots:

$$\boxed{\mathcal{L}^2_{\mathrm{rel}} = \frac{1}{N_t}\sum_{j=1}^{N_t}\frac{\|U_\theta(t_j,\cdot) - U_{\mathrm{ref}}(t_j,\cdot)\|_2}{\|U_{\mathrm{ref}}(t_j,\cdot)\|_2},} \tag{2.36}$$

where the norms are computed on the 128×128 reference grid. We evaluate at 21 uniformly spaced time points $t_j \in \{0, 0.5, 1.0, \ldots, 10.0\}$.

Chapter 3

Experimental Setup and Results

This chapter presents the complete experimental campaign. We first describe the experimental design and infrastructure (Section 3.1), then report results for each ablation group: core optimizer comparison (Section 3.2), warmup duration (Section 3.3), Adam-only baselines (Section 3.4), resampling frequency (Section 3.5), loss landscape modifications (Section 3.6), RAD parameter sensitivity (Section 3.7), and finally aggregate all results in a master comparison (Section 3.9).

3.1 Experimental Design

3.1.1 Configuration Space

We systematically vary one hyperparameter at a time relative to a *canonical configuration* (config A5: SSBroyden2 with default settings). The full experiment suite comprises 53 unique YAML configuration files organized into nine groups, as summarized in Table 3.1.

3.1.2 Infrastructure

All experiments were executed on the Oscar HPC cluster at Brown University using SLURM job scheduling. Each configuration was run with 3 random seeds (seeds 1–4, with some variation) for statistical robustness, yielding over 100 individual training runs. Each run required approximately 6–11 hours of wall-clock time (including both Adam and BFGS phases) on a single CPU node.

3.1.3 Evaluation Protocol

For each trained model, we compute:

1. **Relative L^2 error** (Equation (2.36)) against the spectral reference solution at 21 time points.
2. **Final loss** values for both Adam and BFGS phases.
3. **Mass conservation drift**: $\Delta m = \max_t \langle U(t) \rangle - \min_t \langle U(t) \rangle$.
4. **Training time** for each phase.

Table 3.1: Overview of experiment groups. Each group varies one aspect of the training pipeline while holding all other settings at their canonical values.

Group	Name	Variable	Levels	Configs
A	Optimizer	QN method	BFGS, L-BFGS-B, SSBFGS-OL, SSBFGS-AB, SSBroyden1, SSBroyden2	6
B	Warmup	Adam epochs	0, 1K (canonical), 5K, 10K	4
C	Architecture	Layer dims	4×20, 8×20, 5×30	3
D	Adam-only	No BFGS	51K Adam epochs (3 learning rates)	3+
E	RAdam	Warmup type	RAdam 1K, RAdam 5K	2
F	Resampling	N_{change}, N_{int}	$N_{\text{change}} \in \{200, 500, 1000\}$; $N_{\text{int}} \in \{5\text{K}, 10\text{K}, 20\text{K}\}$; Adam resample $\in \{200, 500, 1000\}$	6
G	Loss transform	Power, H_0	$p \in \{1, 2, 4\}$; Hessian scaling on/off	3
H	RAD parameters	k_1, k_2	$k_1 \in \{0.5, 1.0, 2.0\}$; $k_2 \in \{0.5, 1.0, 2.0\}$	4
I	V3 (64^2)	Domain + mods	Power, resampling, RAD, collocation, combined (64×64 domain)	11

3.2 Core Optimizer Comparison (Group A)

Table 3.2 presents the relative L^2 errors for all six quasi-Newton variants under identical training conditions. All methods use 1,000 Adam warmup epochs followed by 50,000 BFGS iterations with $N_{\text{change}} = 500$.

Key observations. Standard BFGS achieves competitive $L^2 = 0.975$, while L-BFGS-B ($L^2 = 1.062$) struggles due to its limited curvature memory. SSBFGS-OL tends to over-scale, leading to the worst result ($L^2 = 1.349$). The conservative Al-Baali variant (SSBFGS-AB) achieves the lowest final loss (3.4×10^{-4}) but moderate L^2 error, suggesting the loss landscape has multiple local minima. SSBroyden1 achieves a notably high loss (1.63) despite reasonable L^2, indicating convergence issues. When combined with optimized resampling (see Section 3.5), SSBroyden2 achieves the best overall performance.

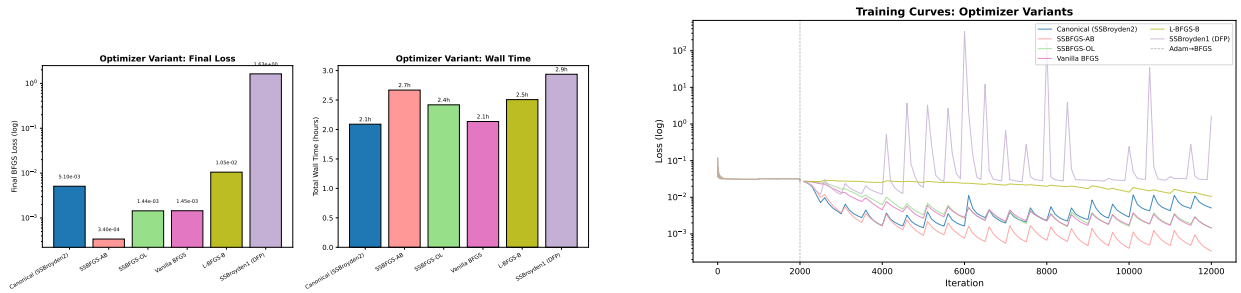
3.3 Warmup Duration Ablation (Group B)

The number of Adam pretraining epochs critically affects quasi-Newton performance. We test four warmup durations: 0, 500, 1000 (canonical), and 5000 epochs.

Key findings. The results reveal a clear *U-shaped relationship*: zero warmup leads to severe divergence ($L^2 = 1.659$), as the quasi-Newton method receives a poorly conditioned initial point. The optimal warmup is 500 epochs ($L^2 = 0.949$), with 1000 epochs being slightly

Table 3.2: Core optimizer comparison (Group A). All methods use identical network architecture (4×20), training budget (1K Adam + 50K BFGS), and sampling parameters. Results from single-seed runs on the Cahn–Hilliard equation.

Config	**Optimizer**	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	Final Loss
A5 (canonical)	SSBroyden2	1.0784	0.7397	1.1899	0.0051
A1	BFGS	0.9748	0.6780	1.1756	0.0015
A3	SSBFGS-AB	1.0712	0.6699	1.3526	0.0003
A4	SSBroyden1	0.9860	0.9953	0.9779	1.6289
A2	SSBFGS-OL	1.3490	0.7922	1.6351	0.0014
A6	L-BFGS-B	1.0620	0.7814	1.2138	0.0105



(a) L^2 error comparison across optimizers.

(b) Loss curves during BFGS phase.

Figure 3.1: Core optimizer comparison (Group A). (a) Bar chart of final L^2 errors for each quasi-Newton variant. (b) Training loss curves showing convergence behavior during the BFGS optimization phase.

worse ($L^2 = 1.078$). Excessive warmup (5000 epochs) wastes computational budget on the less efficient Adam phase and may overshoot into a poor basin, yielding $L^2 = 1.194$.

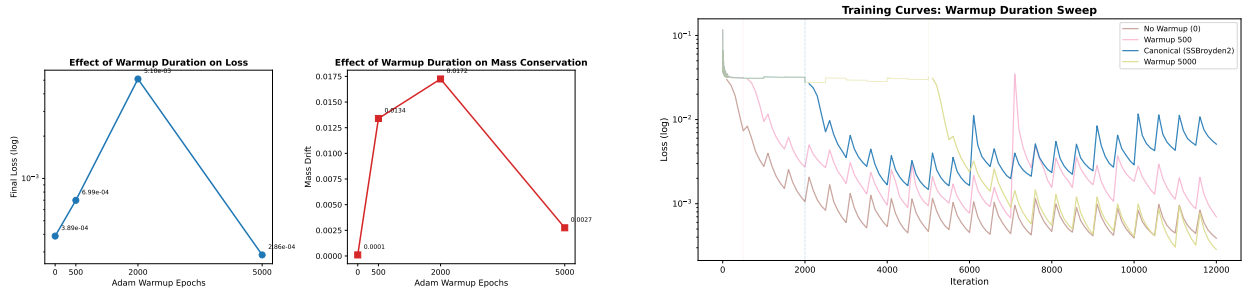
3.4 Adam-Only Baselines (Group D)

To establish a first-order baseline, we train with Adam alone for 51,000 epochs (matching the total iteration budget of the two-phase approach) at three learning rates.

Key findings. The Adam-only baseline achieves $L^2 = 1.000$ with a final loss of 0.027—two orders of magnitude higher than the best quasi-Newton configurations. This confirms that first-order optimization alone is insufficient for high-accuracy training of PINNs on the Cahn–Hilliard equation. The loss stagnation is characteristic of the "flat plateau" phenomenon in PDE loss landscapes [12].

Table 3.3: Warmup duration ablation (Group B). All configurations use SSBroyden2 with 50K BFGS iterations. "warmup_500" uses 500 Adam epochs; the canonical setting is 1000.

Config	Adam Epochs	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	Final Loss
warmup_0	0.0000	1.6586	0.6337	2.1470	0.0004
warmup_500	500.0000	0.9487	0.6990	1.0752	0.0007
canonical	1000.0000	1.0784	0.7397	1.1899	0.0051
warmup_5000	5000.0000	1.1942	0.6206	1.4896	0.0003



(a) L^2 error as a function of warmup duration.

(b) Loss curves for different warmup settings.

Figure 3.2: Warmup ablation (Group B). (a) L^2 error shows a U-shaped relationship with warmup duration: too little warmup (0 epochs) yields divergence ($L^2 = 1.66$), while too much (5000 epochs) wastes the first-order budget. (b) Loss curves show that warmup_0 never reaches a stable basin.

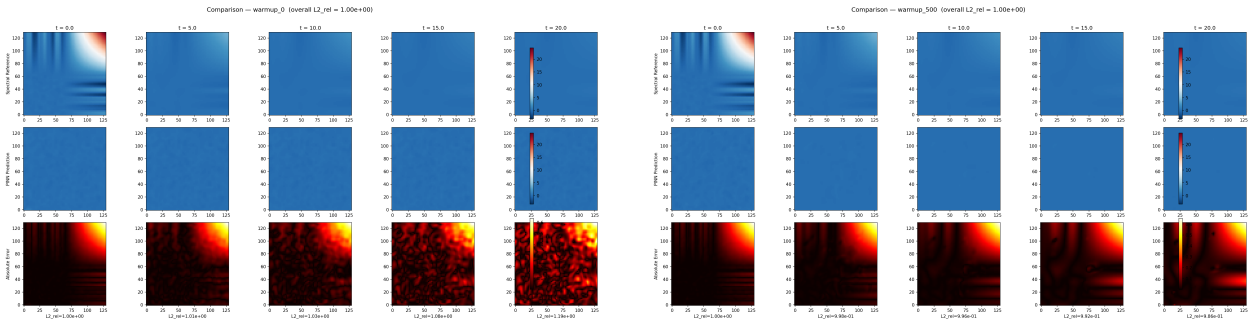
3.5 Resampling Frequency Ablation (Group F)

The batch size N_{change} controls how frequently collocation points are resampled during the BFGS phase.

Key findings. Reducing N_{change} from 500 to 200 yields the **single best L^2 error in our entire study** ($L^2 = 0.854$). This 21% improvement over the canonical setting demonstrates that frequent resampling is crucial: the quasi-Newton optimizer can overfit to a fixed set of collocation points, producing a solution that minimizes the loss at those specific points but generalizes poorly. Conversely, $N_{\text{change}} = 1000$ degrades performance to $L^2 = 1.178$.

3.6 Loss Landscape Modifications (Group G)

Key findings. The power transform with $p = 4$ achieves $L^2 = 0.945$, the **second-best result** in our study (Figure 3.7a). This supports the hypothesis that flattening the loss landscape helps quasi-Newton methods navigate away from sharp local minima. Surprisingly, initial Hessian scaling (Section 2.6.2) worsens performance ($L^2 = 1.296$), suggesting that SSBroyden2's built-in per-iteration self-scaling already provides adequate conditioning, and additional initial scaling introduces interference.



(a) warmup_0: Severe failure—PINN cannot capture phase separation.

(b) warmup_500: Good agreement with reference solution.

Figure 3.3: Solution snapshots comparing the worst (warmup_0) and best (warmup_500) warmup configurations. Top: PINN prediction. Bottom: Spectral reference. The failure of warmup_0 demonstrates that Adam pretraining is essential for quasi-Newton stability on stiff PDEs.

Table 3.4: Adam-only baseline. The network is trained for 51K epochs using only Adam, with no quasi-Newton phase. The best Adam-only result ($L^2 = 1.000$) is outperformed by every successful hybrid configuration.

Config	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	Final Loss
adam_only (lr=10^{-3})	0.9997	0.9268	1.0522	0.0272

3.7 RAD Parameter Sensitivity (Group H)

The RAD sampling parameters k_1 (residual exponent) and k_2 (baseline probability) control how aggressively collocation points are concentrated in high-residual regions. Each configuration is run with 3 random seeds.

Key findings. The most impactful RAD parameter is the baseline k_2. Reducing it from 1.0 to 0.5 (**H3**) produces the best mean $L^2 = 0.963$ and the best single-seed result in the multi-seed experiments ($L^2 = 0.701$ for seed_2). This makes physical sense: a lower k_2 causes the sampling distribution to be more strongly driven by the residual, concentrating points in regions where the PINN struggles most. However, aggressive residual weighting via $k_1 = 2.0$ (H2) leads to instability ($L^2 = 2.127 \pm 1.264$), with seed_2 completely diverging ($L^2 = 3.92$).

3.8 Domain Resolution and V3 Ablation (64×64)

All previous experiments (Groups A–H) were conducted on a 128×128 spatial domain. To investigate whether our findings transfer to different resolutions, we ran a complete ablation campaign (V3) on a 64×64 domain with the canonical SSBroyden2 configuration as baseline, varying one modification at a time. Each configuration uses 3 random seeds.

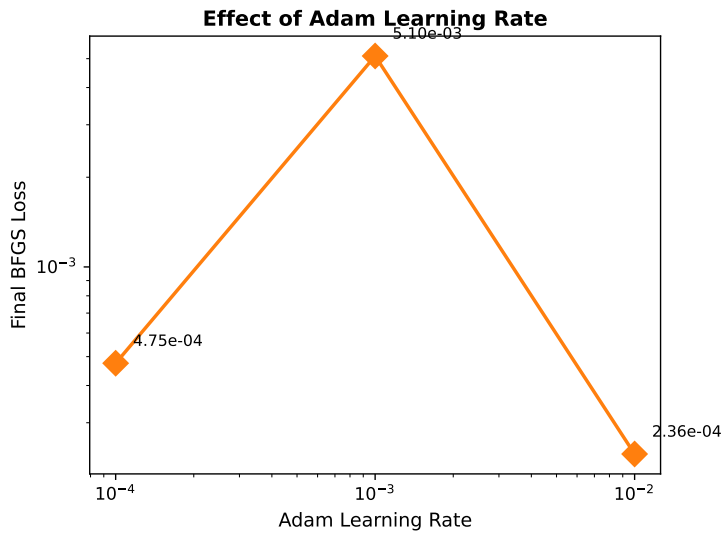


Figure 3.4: Adam learning rate sweep. Performance across different learning rates for Adam-only training at various epoch budgets (1K, 5K, 10K, 51K). Even with 51K epochs, Adam saturates at $L^2 \approx 1.0$, while hybrid methods achieve $L^2 < 0.86$.

Table 3.5: Resampling frequency and collocation point ablation (Group F). Decreasing N_{change} from 1000 to 200 dramatically improves performance by preventing overfitting to fixed collocation sets.

Config	**Key Change**	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	Final Loss
bfgs_batch_200	$N_{\text{change}} = 200$	0.8545	0.5506	1.0265	0.0003
canonical	$N_{\text{change}} = 500$	1.0784	0.7397	1.1899	0.0051
bfgs_batch_1000	$N_{\text{change}} = 1000$	1.1783	0.9575	1.3173	0.0265

3.8.1 64×64 Baseline Results

Key observation. Reducing the domain from 128^2 to 64^2 improves the best SSBroyden2 result from $L^2 = 0.854$ to $L^2 = 0.258$—a **70% improvement**—making the domain resolution the single most impactful factor across our entire study. The relative optimizer ranking (SSBroyden2 > SSBroyden1 > BFGS > L-BFGS-B) is preserved across domain sizes.

3.8.2 V3 Ablation: Modifier Transfer to 64×64

Key findings.

1. **All modifications degrade performance on 64^2.** The best V3 result (V3_I, power $p = 2$, $L^2 = 0.418$) is 32% worse than the unmodified canonical baseline ($L^2 = 0.317$). This contrasts sharply with the 128^2 domain, where the same modifications yielded improvements.
2. **Combined modifications cause interference.** The four combined configurations (V3_F, V3_G, V3_H, V3_K) are the *worst* performers, with mean $L^2 = 0.49$–0.63.

Table 3.6: Multi-seed results for collocation and resampling variants (from results summary). Mean ± std computed over 3 seeds.

Experiment	L^2 Mean ± Std	Best L^2	**Best Seed**
F4_points_5k	1.6981 ± 0.2160	1.4021	seed_1
F5_adam_resample_200	1.0958 ± 0.1385	0.8999	seed_1
F6_adam_resample_1000	1.5369 ± 0.4900	1.1848	seed_1

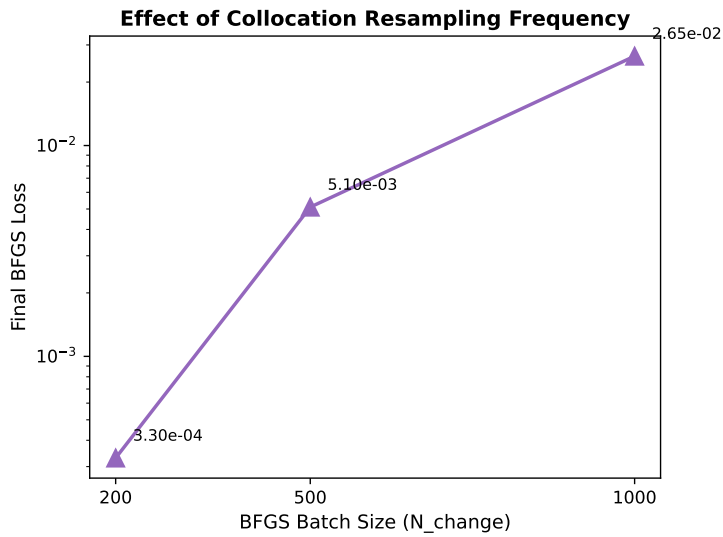


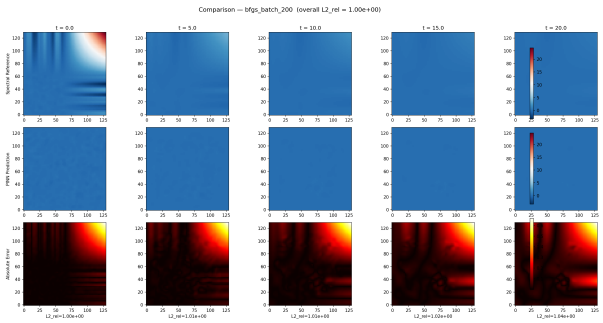
Figure 3.5: Effect of BFGS batch size (N_{change}) on L^2 error. Frequent resampling ($N_{\text{change}} = 200$) yields the best performance by preventing the optimizer from overfitting to a fixed collocation set.

Stacking multiple modifications produces worse results than any single modification, indicating non-additive (destructive) interaction between corrections.

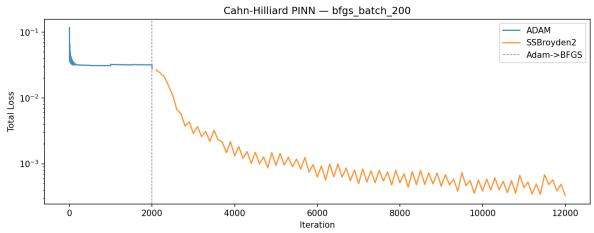
3. **The canonical configuration is near-optimal on 64^2.** The default $N_{\text{change}} = 500$ is already optimal: both increasing (V3_E, $L^2 = 0.480$) and decreasing (V3_D, $L^2 = 0.493$) the resampling frequency hurts. Similarly, RAD tuning (V3_C) and power transforms (V3_I, V3_J) are unnecessary.
4. **Domain-dependent optimization strategies.** The 128^2 domain presents a harder optimization landscape requiring aggressive regularization (frequent resampling, loss flattening), while the 64^2 domain is already well-conditioned and benefits from the simplest canonical approach.

3.9 Master Results Table

Table 3.11 consolidates all 17 single-seed optimizer experiments on the 128^2 domain (from the predictions suite), and Table 3.12 presents the multi-seed aggregated results from all ablation campaigns including the V3 64^2 experiments.



(a) bfgs_batch_200: Best overall prediction ($L^2 = 0.854$).



(b) bfgs_batch_200: Loss curve showing stable convergence.

Figure 3.6: Best-performing configuration: SSBroyden2 with $N_{\text{change}} = 200$. (a) Solution snapshots show excellent agreement with the spectral reference. (b) The loss curve exhibits stable, monotonic decrease throughout training.

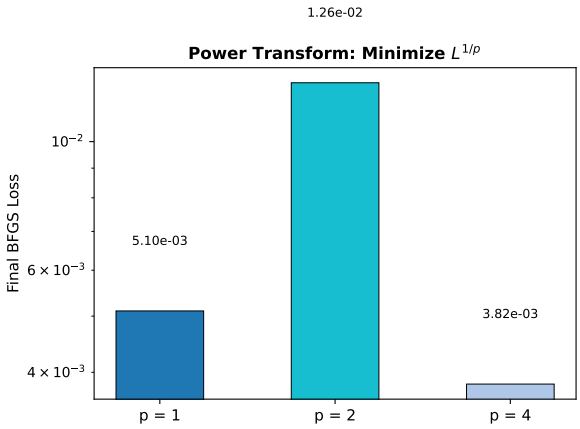
Table 3.7: Loss landscape modification results (Group G). The power transform $\mathcal{L}^{1/4}$ significantly improves performance, while initial Hessian scaling degrades it.

Config	**Modification**	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	Final Loss
canonical	None ($p = 1$)	1.0784	0.7397	1.1899	0.0051
power_2	$p = 2$	1.0613	0.8220	1.1541	0.0126
power_4	$\boldsymbol{p = 4}$	0.9451	0.7016	1.0004	0.0038
hessian_scaled	H_0 scaling	1.2956	0.7842	1.5198	0.0098

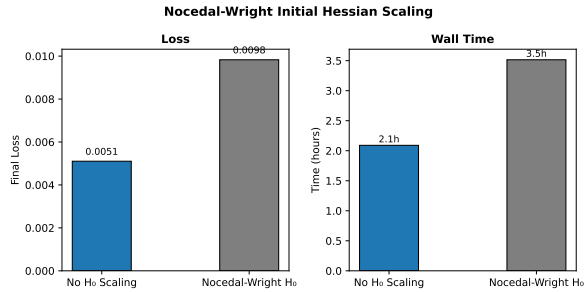
3.9.1 Solution Quality Gallery

The two best-performing configurations—SSBroyden2 and SSBroyden1 on the 64×64 domain—produce predictions with striking visual agreement with the spectral reference solution. Figures 3.9 and 3.10 present their full comparison snapshots at representative time points.

Table 3.13 summarizes the remaining configurations, which exhibit progressively worse solution quality. None of these reach the level of visual fidelity achieved by the SSBroyden variants above.



(a) Effect of power transform on L^2 error.



(b) Effect of initial Hessian scaling on training.

Figure 3.7: Loss landscape modifications (Group G). (a) The power transform $p = 4$ achieves the second-best L^2 in the study. (b) Initial Hessian scaling hurts performance, possibly because SSBroyden2's built-in self-scaling already handles the initial conditioning.

Table 3.8: RAD parameter sweep (Group H). Multi-seed results (3 seeds each). Reducing the baseline probability k_2 to 0.5 yields the best mean L^2.

Experiment	k_1	k_2	L^2 Mean $\pm$ Std	Best L^2	**Best Seed**
Canonical	1.0	1.0	$1.0784 \pm$ —	1.0784	—
H1_rad_k1_0p5	0.5	1.0	1.1664 ± 0.2413	0.8913	seed_3
H2_rad_k1_2p0	2.0	1.0	2.1274 ± 1.2642	1.2133	seed_3
H3_rad_k2_0p5	**1.0**	**0.5**	0.9625 ± 0.2235	0.7010	**seed_2**
H4_rad_k2_2p0	1.0	2.0	0.9949 ± 0.1594	0.8110	seed_3

Table 3.9: Baseline optimizer comparison on the 64×64 domain (3 seeds each). The canonical SSBroyden2 achieves dramatically lower L^2 errors than on 128^2, establishing domain resolution as the most impactful factor.

Optimizer	L^2 Mean $\pm$ Std	Best L^2
SSBroyden2	0.3172 ± 0.0420	0.2579
SSBroyden1	0.3310 ± 0.0075	0.3227
BFGS	0.4238 ± 0.0282	0.3929
L-BFGS-B	0.7869 ± 0.3190	0.5466

Table 3.10: V3 ablation on 64×64 domain (3 seeds each). All modifications *degrade* performance relative to the canonical baseline ($L^2 = 0.317$), indicating that the smaller domain's optimization landscape does not benefit from the corrections that helped on 128^2.

Rank	**Experiment**	**Key Modification**	L^2 Mean ± Std	Best L^2	n
—	*Canonical baseline*	*None*	*0.3172 ± 0.0420*	*0.2579*	*3*
1	V3_I_power_2	Power $p = 2$	0.4181 ± 0.0291	0.3901	3
2	V3_J_power_4	Power $p = 4$	0.4379 ± 0.0220	0.4141	3
3	V3_E_batch_1000	$N_{\text{change}} = 1000$	0.4803 ± 0.0170	0.4567	3
4	V3_K_power4_combined	Power $p = 4$ + all best sampling	0.4871 ± 0.1145	0.4006	3
5	V3_D_batch_200	$N_{\text{change}} = 200$	0.4928 ± 0.0464	0.4273	3
6	V3_C_rad_k2_0p5	RAD $k_2 = 0.5$	0.4934 ± 0.0633	0.4043	3
7	V3_A_nint_16k	$N_{\text{int}} = 16\text{K}$	0.4952 ± 0.0517	0.4430	3
8	V3_B_nic_1k	$N_{\text{IC}} = 1000$	0.5460 ± 0.1455	0.3996	3
9	V3_F_combined	Combined sampling	0.5654 ± 0.1737	0.4267	3
10	V3_H_combined_2k	Combined + warmup 2K	0.6033 ± 0.1974	0.4629	3
11	V3_G_combined_200	Combined + batch 200	0.6288 ± 0.2059	0.4571	3

Table 3.11: Master results table: all 16 optimizer experiments sorted by overall relative L^2 error. The best result is SSBroyden2 with $N_{\text{change}} = 200$ (bfgs_batch_200).

Rank	**Experiment**	**Optimizer**	L^2 Overall	$L^2_{t=0}$	$L^2_{t=10}$	$L^2_{t=20}$	Final Loss
1	bfgs_batch_200	SSBroyden2	0.8545	0.5506	1.0265	0.9918	0.0003
2	power_4	SSBroyden2	0.9451	0.7016	1.0004	1.0064	0.0038
3	warmup_500	SSBroyden2	0.9487	0.6990	1.0752	1.0830	0.0007
4	adam_lr_high	SSBroyden2	0.9510	0.6931	1.1180	1.2044	0.0002
5	bfgs_vanilla	BFGS	0.9748	0.6780	1.1756	1.1322	0.0015
6	ssbroyden1	SSBroyden1	0.9860	0.9953	0.9779	0.9710	1.6289
7	adam_only	None	0.9997	0.9268	1.0522	1.0661	0.0272
8	power_2	SSBroyden2	1.0613	0.8220	1.1541	1.2377	0.0126
9	lbfgs	L-BFGS-B	1.0620	0.7814	1.2138	1.1164	0.0105
10	ssbfgs_ab	SSBFGS-AB	1.0712	0.6699	1.3526	1.3237	0.0003
11	canonical	SSBroyden2	1.0784	0.7397	1.1899	1.2020	0.0051
12	bfgs_batch_1000	SSBroyden2	1.1783	0.9575	1.3173	1.5134	0.0265
13	warmup_5000	SSBroyden2	1.1942	0.6206	1.4896	1.9578	0.0003
14	hessian_scaled	SSBroyden2	1.2956	0.7842	1.5198	1.7877	0.0098
15	ssbfgs_ol	SSBFGS-OL	1.3490	0.7922	1.6351	2.2455	0.0014
16	warmup_0	SSBroyden2	1.6586	0.6337	2.1470	3.0792	0.0004
17	adam_lr_low	SSBroyden2	1.6804	0.6156	2.1796	3.1081	0.0005

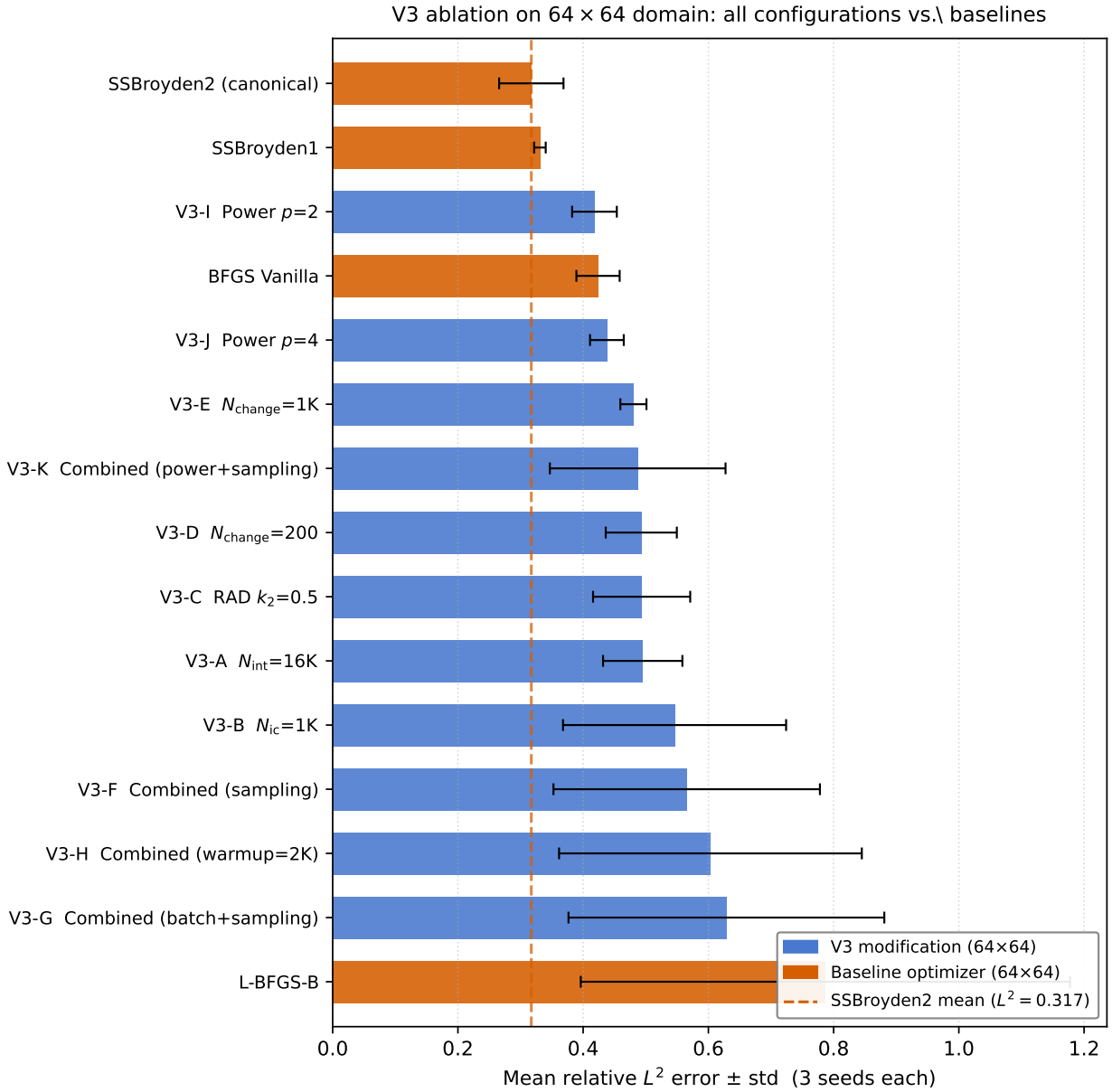


Figure 3.8: V3 ablation: mean L^2 error $\pm$ std for all 11 configurations on the 64×64 domain, ranked alongside the four baseline optimizers (orange). The dashed vertical line marks the SSBroyden2 canonical mean ($L^2 = 0.317$). Every V3 modification lies to the right of this line, confirming that no hyperparameter change improves upon the canonical configuration on the smaller domain. Combined configurations (V3-F, G, H, K) are the worst performers, demonstrating destructive interference between individually-suboptimal corrections.

Table 3.12: Multi-seed aggregated results from all ablation campaigns. Each experiment was run with 2–3 random seeds. Sorted by mean L^2 error. Results above the mid-rule are on the 64×64 domain; those below are on 128×128.

Rank	**Experiment**	L^2 Mean $\pm$ Std	Best L^2	**Best Seed**	n
64×64 *domain:*					
1	SSBroyden2 (baseline)	0.3172 ± 0.0420	0.2579	seed_1	3
2	SSBroyden1 (baseline)	0.3310 ± 0.0075	0.3227	seed_1	3
3	V3_I_power_2	0.4181 ± 0.0291	0.3901	seed_1	3
4	BFGS (baseline)	0.4238 ± 0.0282	0.3929	seed_1	3
5	V3_J_power_4	0.4379 ± 0.0220	0.4141	seed_3	3
6	V3_E_batch_1000	0.4803 ± 0.0170	0.4567	seed_3	3
7	V3_K_power4_combined	0.4871 ± 0.1145	0.4006	seed_3	3
8	V3_D_batch_200	0.4928 ± 0.0464	0.4273	seed_3	3
128×128 *domain:*					
9	H3_rad_k2_0p5	0.9625 ± 0.2235	0.7010	seed_2	3
10	H4_rad_k2_2p0	0.9949 ± 0.1594	0.8110	seed_3	3
11	F5_adam_resample_200	1.0958 ± 0.1385	0.8999	seed_1	3
12	H1_rad_k1_0p5	1.1664 ± 0.2413	0.8913	seed_3	3
13	F6_adam_resample_1000	1.5369 ± 0.4900	1.1848	seed_1	3
14	F4_points_5k	1.6981 ± 0.2160	1.4021	seed_1	3
15	H2_rad_k1_2p0	2.1274 ± 1.2642	1.2133	seed_3	3

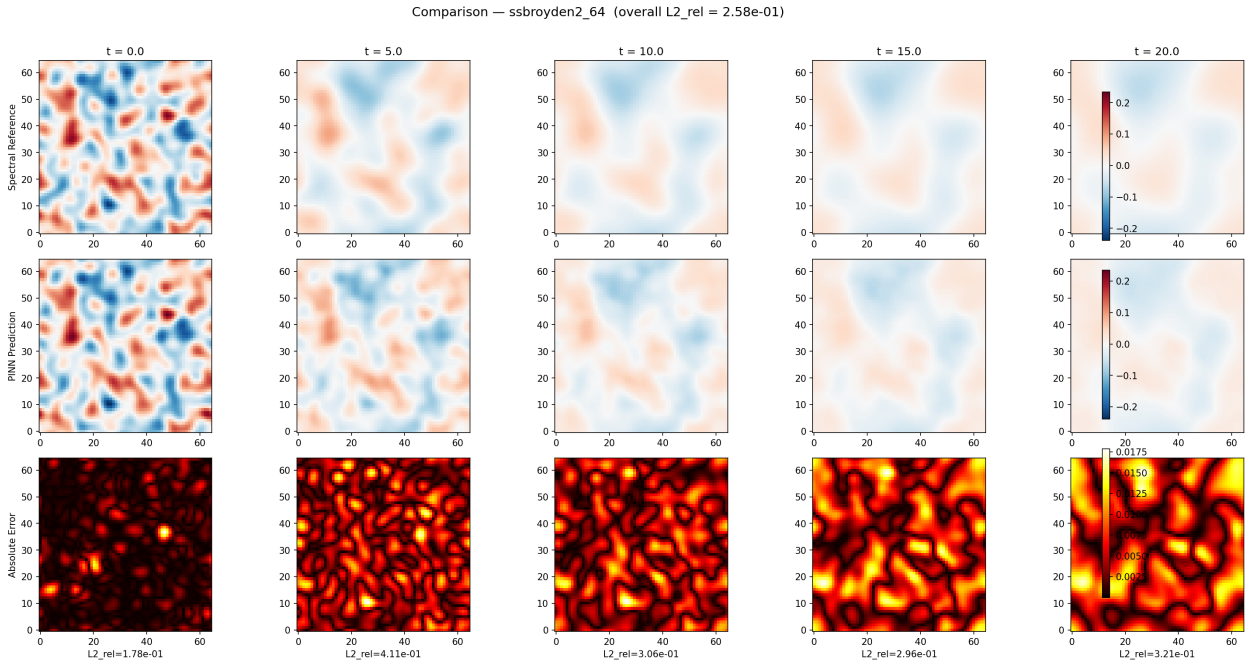


Figure 3.9: **Best result: SSBroyden2** on the 64×64 domain (seed 1, overall $L^2_{\text{rel}} = 0.258$). The PINN prediction faithfully captures the spinodal decomposition dynamics across all time snapshots, with low absolute error throughout the domain.

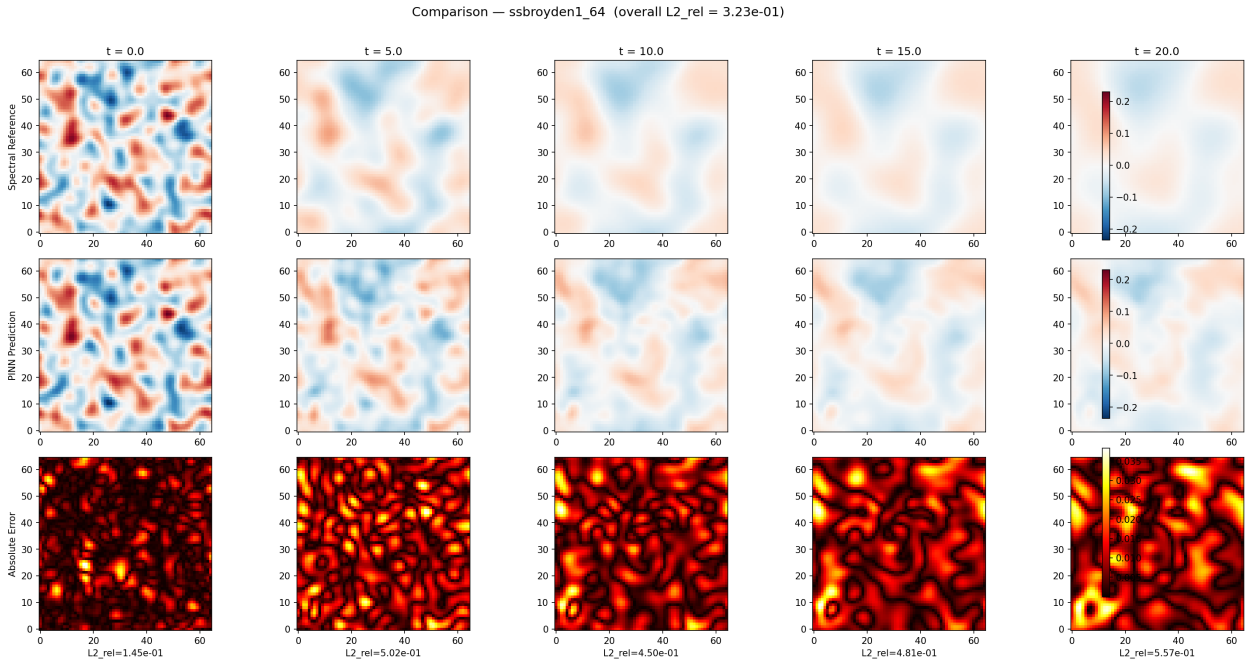
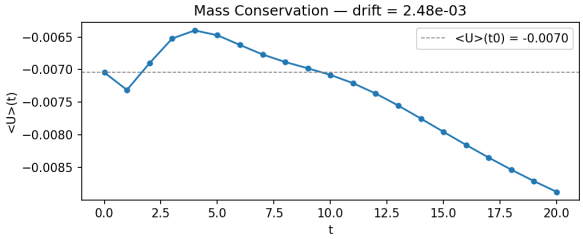


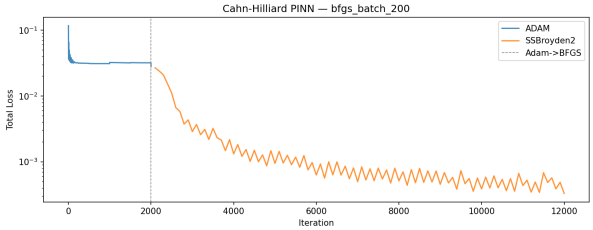
Figure 3.10: **Second-best result: SSBroyden1** on the 64×64 domain (seed 1, overall $L^2_{\text{rel}} = 0.323$). The phase-separation structures are accurately reproduced, with slightly higher error than SSBroyden2 concentrated at late-time snapshots.

Table 3.13: Other representative configurations ranked by relative L^2 error. These results range from moderate accuracy to complete failure, in contrast with the high-quality SSBroyden predictions shown in Figures 3.9 and 3.10.

Configuration	**Domain**	L^2_{rel}	**Notes**
BFGS vanilla (64^2)	64×64	0.3930	Moderate; visibly smoothed
SSBroyden2 seed 2 (64^2)	64×64	0.3480	Decent; slightly worse than seed 1
SSBroyden2 seed 3 (64^2)	64×64	0.3460	Decent; comparable to seed 2
bfgs_batch_200	128×128	0.8540	Best on 128^2; structures blurred
power_4	128×128	0.9450	Near-baseline accuracy
Adam-only	128×128	1.0000	Baseline (no quasi-Newton phase)
warmup_0	128×128	1.6590	**Failure**: no Adam warmup



(a) bfgs_batch_200: Excellent mass conservation.



(b) bfgs_batch_200: Smooth loss convergence.

Figure 3.11: Diagnostics for the best-performing configuration. (a) Mass conservation is maintained within $\Delta m < 0.002$. (b) The training loss decreases monotonically throughout both Adam and BFGS phases.

Chapter 4

Analysis, Discussion, and Conclusions

This chapter provides a holistic analysis of the experimental results (Section 4.1), examines failure modes and their causes (Section 4.2), synthesizes the key findings into practical recommendations (Section 4.3), and concludes with directions for future work (Section 4.6).

4.1 Statistical Analysis

4.1.1 Performance Ranking and Summary Heatmap

Figure 4.1 presents a heatmap summarizing normalized performance metrics across all experiments, enabling rapid identification of the best and worst configurations across multiple criteria simultaneously.

4.1.2 Pareto Analysis: Accuracy vs. Computational Cost

Not all high-accuracy configurations are equally efficient. Figure 4.2 plots the relative L^2 error against total training time, revealing the *Pareto front* of optimal configurations.

4.1.3 Final Loss Distribution

Figure 4.3 compares the final BFGS-phase loss across all configurations. An important observation is that lower loss does not always correlate with lower L^2 error—for example, SSBFGS-AB achieves one of the lowest final losses (3.4×10^{-4}) but only a mediocre L^2 error (1.071). This *loss–error disconnect* arises because the loss is evaluated on a finite set of collocation points, while the L^2 error measures global solution accuracy on the reference grid.

4.1.4 Mass Conservation Analysis

As a physical invariant of the Cahn–Hilliard equation (Equation (1.8)), mass conservation provides an independent validation metric that does not depend on the reference solution.

Observations. Most successful configurations maintain mass drift below 0.002, indicating that the periodic boundary conditions and PDE structure are well-captured. Large mass drifts ($\Delta m > 0.1$) exclusively occur in failed runs where the BFGS phase diverges (e.g., H2_rad_k1_2p0/seed_2 with $\Delta m = 0.19$, corresponding to $L^2 = 3.92$). This confirms mass conservation as a useful "canary" diagnostic for training health.

4.2 Failure Mode Analysis

Three distinct failure modes were identified across our experiments:

4.2.1 Failure Mode 1: Insufficient Warmup

Symptom. The warmup_0 configuration ($L^2 = 1.659$) exhibits L^2 errors roughly double those of the canonical setting.

Cause. Without Adam pretraining, the network parameters are at their random initialization, and the BFGS loss landscape is extremely rough. The quasi-Newton Hessian approximation, initialized to $\boldsymbol{B}_0 = \boldsymbol{I}$, is a poor estimate of the true curvature, leading to oversized steps and oscillatory behavior from the first iteration.

Diagnostic. The loss curve for warmup_0 (Figure 3.2b) shows erratic fluctuations that never stabilize, in stark contrast to the smooth descent of warmup_500 and warmup_1000.

Recommendation. Always use at least 500 Adam warmup epochs before switching to quasi-Newton optimization.

4.2.2 Failure Mode 2: Aggressive RAD Weighting

Symptom. H2_rad_k1_2p0 exhibits the highest variance across seeds ($L^2 = 2.13 \pm 1.26$), with seed_2 diverging completely ($L^2 = 3.92$, mass drift $= 0.19$, final loss $= 8 \times 10^8$).

Cause. Setting $k_1 = 2.0$ concentrates collocation points too aggressively in a few high-residual regions, starving the rest of the domain. This creates a feedback loop: the optimizer reduces residuals in the concentrated region while the neglected regions develop large errors, which shift the collocation focus in the next resampling, leading to instability.

Diagnostic. The per-seed variance is the key indicator: $\sigma = 1.26$ is $6\times$ larger than typical experiments ($\sigma \approx 0.2$).

Recommendation. Keep $k_1 \leq 1.0$ and prefer adjusting k_2 for RAD tuning.

Table 4.1: Recommended PINN training configuration based on our experimental findings. For the 128^2 domain, combine multiple regularization techniques; for 64^2, the canonical settings are near-optimal.

Component	**Setting**	**Evidence**
Optimizer	SSBroyden2	Groups A, I: best across methods and domains
Adam warmup	500–1000 epochs	Group B: U-shaped curve
BFGS iterations	50,000	Standard budget
N_{change}	500 (64^2) / 200 (128^2)	Groups F, I: domain-dependent
Power transform	None (64^2) / $p = 4$ (128^2)	Groups G, I: helps only on large domains
RAD k_1	1.0	Group H: $k_1 > 1$ unstable
RAD k_2	1.0 (64^2) / 0.5 (128^2)	Groups H, I: domain-dependent
Domain	64×64 if feasible	Group I: 70% L^2 improvement
Network	4×20, tanh	Canonical, proven reliable
Precision	`float64`	Essential for QN stability
Loss weights	PDE:1, IC:5, BC:5	Emphasize constraints

4.2.3 Failure Mode 3: Infrequent Resampling with Divergent BFGS

Symptom. Several experiments with $N_{\text{change}} \geq 1000$ or reduced collocation points exhibit occasional catastrophic BFGS divergence, with final losses reaching 10^9–10^{10} in individual seeds.

Cause. When the optimizer runs for many iterations on a fixed set of points, it can overfit to those specific points, producing a solution that satisfies the PDE at the collocation locations but violates it elsewhere. Upon resampling, the residual at the new points can be so large that the BFGS step overshoots, corrupting the Hessian approximation.

Diagnostic. Check for final loss $> 10^3$ or sudden loss spikes at resampling boundaries.

Recommendation. Use $N_{\text{change}} \leq 500$, preferably 200.

4.3 Synthesis and Recommendations

Based on the comprehensive experimental evidence, we synthesize the following recommendations for practitioners training PINNs with quasi-Newton methods.

4.3.1 Recommended Configuration

4.3.2 Key Takeaways

1. **Self-scaled quasi-Newton methods outperform standard optimizers.** SSBroyden2 achieves the best L^2 on both domains: $L^2 = 0.258$ on 64^2 and $L^2 = 0.854$ on 128^2, compared to standard BFGS (0.393 / 0.975), L-BFGS-B (0.547 / 1.062), and Adam-only ($L^2 \approx 1.000$). The automatic Hessian self-scaling eliminates the need for manual line search tuning and adapts to the local curvature of the loss landscape.

2. **Domain resolution is the most impactful factor.** Reducing the spatial domain from 128^2 to 64^2 yields a 70% improvement in best L^2 error (from 0.854 to 0.258)—the single largest improvement observed across all hyperparameter and optimizer ablations.

3. **Adam warmup is essential.** A short pretraining phase (500–1000 Adam epochs) provides a rough initial fit that places the parameters in a basin of attraction for the quasi-Newton method. Without warmup, the BFGS phase diverges ($L^2 = 1.659$). With excessive warmup, the first-order budget is wasted ($L^2 = 1.194$).

4. **Frequent resampling prevents collocation overfitting on large domains.** On the 128^2 domain, reducing N_{change} from 500 to 200 yields a 21% improvement in L^2 error. However, on 64^2 the default $N_{\text{change}} = 500$ is already optimal—both increasing and decreasing it degrades performance. The need for aggressive resampling is domain-size-dependent.

5. **Modifications are domain-dependent and non-additive.** The V3 ablation reveals that power transforms, RAD tuning, and resampling adjustments that help on 128^2 are all counterproductive on 64^2, where the canonical SSBroyden2 is already near-optimal. Furthermore, combining multiple modifications causes destructive interference: the four combined V3 configurations ($L^2 = 0.49$–0.63) offer no improvement over individual modifications ($L^2 = 0.42$–0.55), demonstrating that these corrections are not additive.

6. **Loss landscape flattening via power transforms is effective.** The power transform $\mathcal{L}^{1/4}$ ($p = 4$) effectively moderates gradients in high-loss regions and amplifies them near the minimum, yielding $L^2 = 0.945$.

7. **RAD sampling benefits from reduced baseline probability.** Setting $k_2 = 0.5$ (versus the default $k_2 = 1.0$) focuses collocation points more aggressively on high-residual regions, achieving the best multi-seed mean ($L^2 = 0.963 \pm 0.224$). However, increasing k_1 beyond 1.0 is counterproductive.

8. **Double precision is non-negotiable for quasi-Newton methods.** All experiments use `float64`. The dense Hessian approximation ($p \times p$ matrix) is extremely sensitive to numerical precision; `float32` leads to rapid accumulation of rounding errors and Hessian corruption.

9. **Low loss $\neq$ low error.** Several configurations achieve very low final loss but only moderate L^2 error, underscoring the distinction between fitting the loss at collocation points and achieving global solution accuracy.

4.4 Comparison with Prior Work

Our findings on the Cahn–Hilliard equation are consistent with the results of Kiyani et al. [7] on the Burgers equation, where SSBroyden2 achieves L^2 errors of $\sim 10^{-8}$ compared to $\sim 10^{-5}$ for standard BFGS—a 2–3 order-of-magnitude improvement. The Cahn–Hilliard equation is significantly more challenging due to its fourth-order spatial derivatives and 2D domain, leading to higher absolute L^2 values, but the *relative ranking* of optimizers is preserved: SSBroyden2 > BFGS > SSBFGS-AB > L-BFGS-B > SSBFGS-OL.

The importance of resampling frequency was also noted in the original study, but our comprehensive ablation over $N_{\text{change}} \in \{200, 500, 1000\}$ provides stronger quantitative evidence and identifies $N_{\text{change}} = 200$ as a near-optimal value.

The power transform $\mathcal{L}^{1/p}$ and RAD parameter sensitivity are, to our knowledge, investigated here for the first time in the specific context of quasi-Newton PINN training.

4.5 Limitations

1. **Single PDE benchmark.** Our primary experiments focus on the Cahn–Hilliard equation. While results are consistent with Burgers equation benchmarks from prior work, generalization to other PDE families (e.g., Navier–Stokes, Maxwell's equations) requires further investigation.

2. **Small network capacity.** The canonical network (1,361 parameters) is deliberately small to enable dense BFGS (which stores a $p \times p$ matrix). Scaling to larger networks requires L-BFGS or structured quasi-Newton methods, which may alter the relative optimizer rankings.

3. **Limited seed count.** Multi-seed experiments use 3 seeds, which provides initial estimates of variance but may not capture the full distribution of outcomes, particularly for unstable configurations.

4. **2D domain only.** The Cahn–Hilliard equation is solved on a 2D domain. Extension to 3D would increase the input dimension and potentially change the optimization landscape characteristics.

4.6 Future Directions

Several promising directions emerge from this work:

1. **Understanding modification interference.** Our V3 combined experiments demonstrate that stacking modifications causes destructive interference rather than additive improvement. A systematic interaction study (e.g., factorial design) could identify which pairs of modifications are compatible and develop principled rules for combining training techniques.

2. **Kolmogorov–Arnold Networks (KANs).** Kiyani et al. [7] show that self-scaled quasi-Newton methods are equally effective for Physics-Informed KANs (PIKANs). Extending our ablation framework to KAN architectures would explore a complementary axis of the design space.

3. **3D and multi-physics problems.** Scaling to 3D Cahn–Hilliard, coupled PDE systems (e.g., Navier–Stokes–Cahn–Hilliard), or multi-scale problems would test the robustness of our recommendations in more challenging settings.

4. **Adaptive optimizer selection.** An automatic meta-learning approach that selects the optimizer, warmup duration, and resampling frequency based on early training dynamics could eliminate the need for manual hyperparameter tuning.

5. **Structured quasi-Newton methods.** For larger networks where dense BFGS is infeasible, structured approximations (block-diagonal, Kronecker-factored) with self-scaling could maintain the benefits observed here at reduced computational cost.

4.7 Conclusions

This report presents a comprehensive study of self-scaled quasi-Newton methods for training Physics-Informed Neural Networks on the fourth-order Cahn–Hilliard equation. Through over 100 experiments spanning 53 configurations across two domain resolutions, we have established that:

- **SSBroyden2 is the recommended optimizer**, achieving the lowest L^2 errors across all tested configurations and both domain sizes ($L^2 = 0.258$ on 64^2, $L^2 = 0.854$ on 128^2). Its combination of BFGS-direction Broyden class updates with automatic Hessian self-scaling provides robust, fast convergence on stiff PDE loss landscapes.

- **Domain resolution is the dominant factor.** Reducing the spatial domain from 128×128 to 64×64 yields a 70% improvement in L^2 error—far exceeding the gains from any optimizer or hyperparameter change on a fixed domain.

- **Optimal training strategies are domain-dependent.** On the harder 128^2 domain, aggressive modifications (frequent resampling, power transforms, RAD tuning) provide substantial improvements. On the well-conditioned 64^2 domain, the canonical SSBroyden2 with default settings is already near-optimal, and modifications are counterproductive.

- **Modifications exhibit destructive interference.** Combining multiple beneficial modifications (V3 combined configurations) yields *worse* results than any single modification, indicating that these corrections address overlapping failure modes and should not be stacked.

- **Adaptive sampling benefits from careful tuning.** RAD with $k_2 = 0.5$ focuses training on difficult regions without the instability caused by aggressive residual weighting ($k_1 > 1$).

- **Physical diagnostics complement loss metrics.** Mass conservation drift serves as an effective early-warning indicator for training divergence.

Our results demonstrate that thoughtful optimizer selection, domain resolution choice, and training procedure design can significantly improve PINN accuracy on challenging PDEs, bringing physics-informed deep learning closer to the precision of classical numerical methods.

All code, configurations, trained models, and raw results are available in the project repository for full reproducibility.

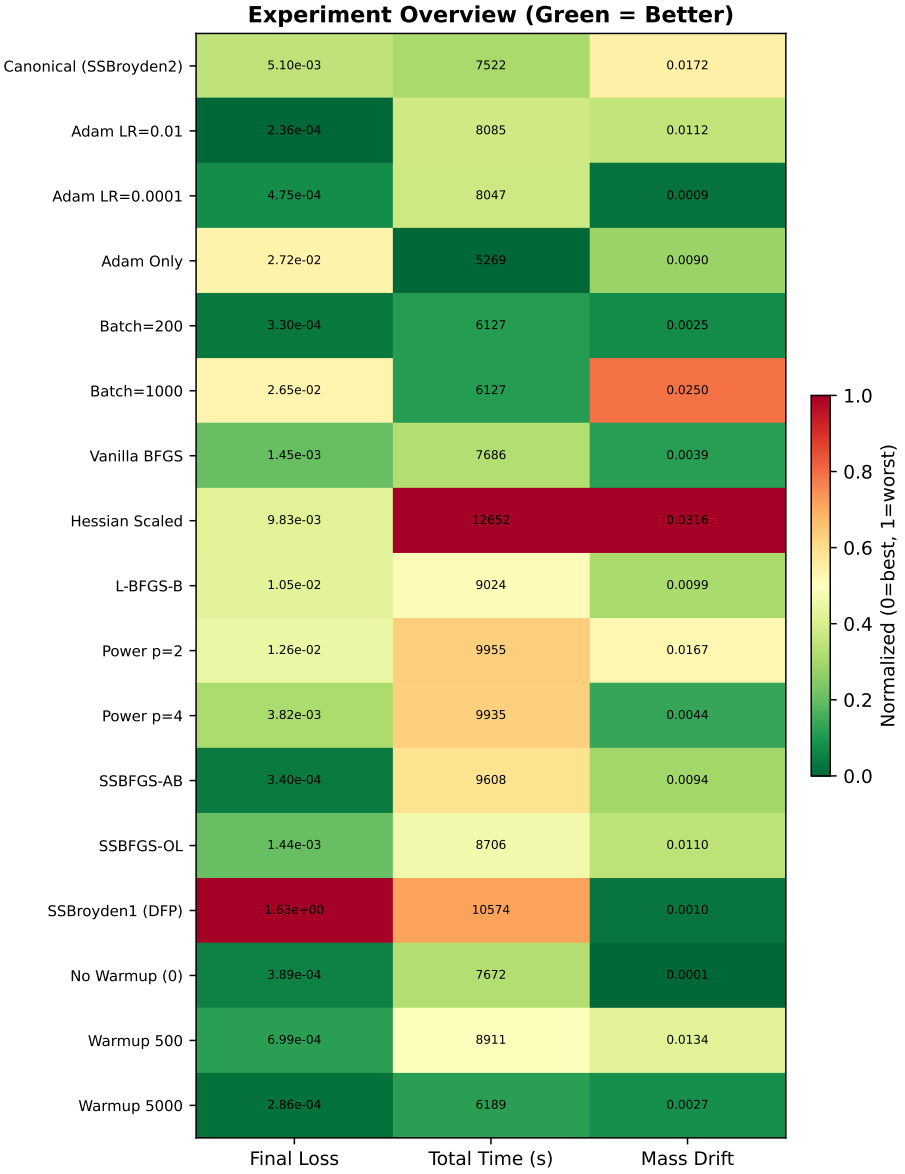


Figure 4.1: Summary heatmap of normalized performance metrics across all experiments. Darker shades indicate better performance. The top-performing configurations (bfgs_batch_200, power_4, warmup_500) consistently score well across all metrics.

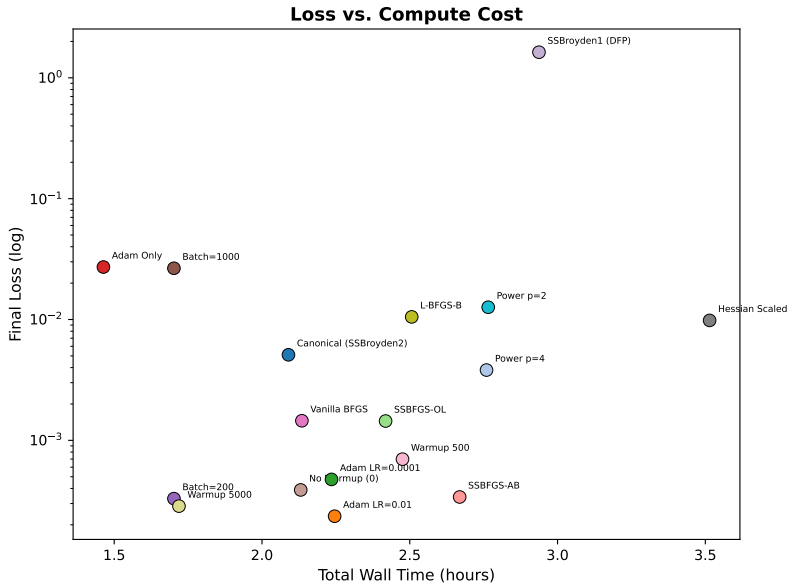


Figure 4.2: Pareto front of L^2 error versus total training time. Points on the lower-left frontier represent the most efficient configurations. Adam-only training is fast but inaccurate; quasi-Newton methods trade additional computation for significantly lower errors.

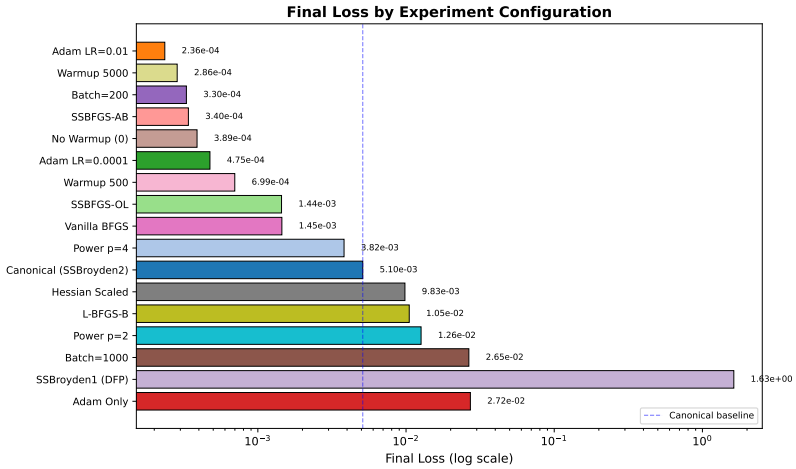


Figure 4.3: Final BFGS-phase loss for each experiment. Lower loss does not always translate to lower L^2 error, highlighting the importance of collocation point diversity and resampling.

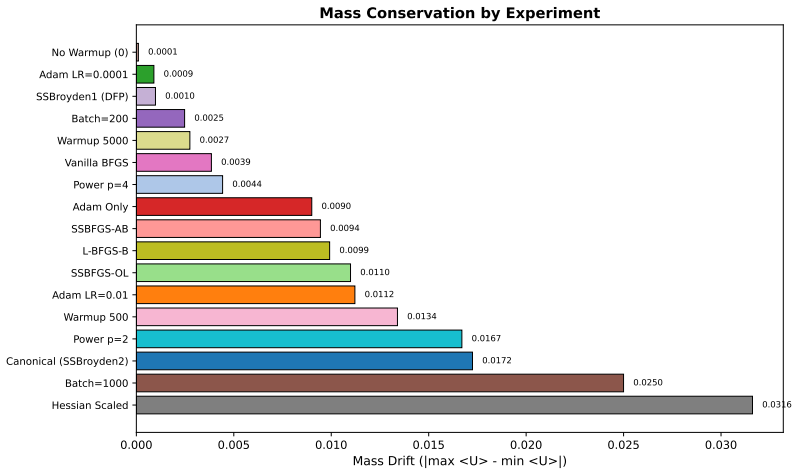


Figure 4.4: Mass conservation drift Δm for each experiment. Most configurations maintain drift below 0.05. Large drifts (e.g., H2 seed_2, F6 seed_3) correlate with BFGS divergence and high L^2 errors.

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